HACK CLUB FALLEN - MASTER GUIDE PART 1: HAPTIC NAVIGATION CORE

Section 1: Haptic Navigation System (2-Hour Build Guide)

Neckband Ultrasonic Array, Sleeve Box Electronics (BNO085) & Stitched Haptics

1. System Parts & Hardware Components

Component	Qty	Function / Purpose
ESP32 Microcontroller (#1)	1	Main sleeve-box MCU handling ultrasonic processing & haptics.
BNO085 9-Axis Compass Module	1	Inside sleeve box for absolute orientation & spatial matrix calculation.
HC-SR04 Ultrasonic Sensors	4	Mounted on Neckband (Front, Rear, Left, Right obstacle scan).
Coin Vibration Motors	4	Stitched inside fabric sleeve lining for 360° directional feedback.
S8050 NPN Transistors	4	Switching motor drivers triggered by ESP32 PWM pins.

2. Hard-Wired Circuit Connections

ESP32 Pin	Connected Component	Wiring Details
GPIO 21 (SDA) / 22 (SCL)	BNO085 Compass Module	I2C Bus inside sleeve box with 4.7kΩ pull-up resistors.
GPIO 4, 5, 16, 17 (Trig) / 18, 19, 23, 25 (Echo)	4x Neckband HC-SR04 Sensors	Hard-wired cable harness running down shoulder to sleeve box.
GPIO 12, 13, 14, 15	4x Vibration Motors (N, S, E, W)	Connected to base of S8050 NPN transistors via 1kΩ resistors.

3. 2-Hour Assembly & Stitching Instructions

1. **Neckband Harness:** Mount 4 HC-SR04 sensors facing 4 cardinal directions on a comfortable soft neckband. Route signals into a braided sleeve cable down to the arm.
2. **Sleeve Control Box:** Mount ESP32 #1 and BNO085 module inside a compact 3D-printed palm/wrist box attached to a velcro sleeve mount.
3. **Fabric Haptic Integration:** Hand-stitch 4 coin vibration motors into the sleeve's inner fabric around the forearm to deliver clear tactile feedback.

4. Complete Haptic Device Code (BNO085 + Neckband Ultrasonics)

```
#include <Wire.h> #include <Adafruit_BNO08x.h> Adafruit_BNO08x bno08x; sh2_SensorValue_t sensorValue; // Neckband U ltrasonic Pins (Front, Rear, Left, Right) const int TRIG_PINS[4] = {4, 5, 16, 17}; const int ECHO_PINS[4] = {18, 1 9, 23, 25}; const int MOTOR_PINS[4] = {12, 13, 14, 15}; long getDistance(int trig, int echo) { digitalWrite(trig, L OW); delayMicroseconds(2); digitalWrite(trig, HIGH); delayMicroseconds(10); digitalWrite(trig, LOW); long duration = pulseIn(echo, HIGH, 25000); // 25ms timeout return (duration == 0) ? 999 : (duration * 0.034 / 2); } void setup() { Wire.begin(21, 22); bno08x.begin_I2C(); bno08x.enableReport(SH2_ARVR_STABILIZED_RV); for(int i=0; i<4; i++) { pin Mode(TRIG_PINS[i], OUTPUT); pinMode(ECHO_PINS[i], INPUT); pinMode(MOTOR_PINS[i], OUTPUT); } } void loop() { if (bno 08x.getSensorEvent(&sensorValue)) { for(int i=0; i<4; i++) { long dist = getDistance(TRIG_PINS[i], ECHO_PINS[i]); i f (dist < 50) { // Obstacle closer than 50cm analogWrite(MOTOR_PINS[i], map(dist, 5, 50, 255, 100)); } else { analo gWrite(MOTOR_PINS[i], 0); } } } delay(30); }
```

5. Building Rules: DO's and DON'Ts

DO's

- Use 1N4007 flyback diodes across vibration motor terminals.
- Calibrate BNO085 compass module away from high-current battery leads.
- Strain-relief the neckband cable harness at the shoulder junction.

DON'T's

- DO NOT connect Iron Man MPU6050 data to the haptic system.
- Do not leave ultrasonic pulseIn calls without a microsecond timeout.

PART 2: MECHANICAL IRON MAN GAUNTLET

Section 2: Iron Man Gauntlet (Mechanical Focus)

100% Mechanical Armor Mechanics & Ultra-Simple MPU6050 Orientation Tracking

1. Mechanical Rig Architecture & Electronics

Component	Qty	Function & Specifications
3D Printed Expanding Linkages	1 set	Forearm & wrist mechanical ridge shell (Mechanical focus).
Elastic Tension Bands + Velcro	4	Snaps gauntlet onto arm instantly with single-tab pulling action.
ESP32 Microcontroller (#2)	1	Dedicated MCU reading MPU6050 motion (Zero complex peripherals).
MPU6050 6-Axis IMU	1	Mounted on gauntlet to record raw wrist pitch, roll & acceleration.

2. Simplified Motion Code (ESP32 #2 + MPU6050)

No NeoPixels, no servos, no bloated drivers. Pure, bulletproof motion tracking:

```
#include <Wire.h> #include <Adafruit_MPU6050.h> Adafruit_MPU6050 mpu; void setup() { Serial.begin(115200); Wire.beg
in(21, 22); mpu.begin(); mpu.setAccelerometerRange(MPU6050_RANGE_8_G); mpu.setGyroRange(MPU6050_RANGE_500_DEG); } v
oid loop() { sensors_event_t a, g, temp; mpu.getEvent(&a, &g, &temp); // Stream raw orientation data for gesture tr
acking / telemetry float pitch = a.acceleration.x; float roll = a.acceleration.y; if (pitch > 6.0) { Serial.println
("GAUNTLET_GESTURE: REPULSOR_RAISED"); } delay(20); }
```

3. Mechanical Assembly Technique

- **Linkage Ridges:** Assemble 3D-printed ridges with pivot pins. Run high-tensile elastic bands underneath to create an auto-expanding tensioner.
- **Hand Mount:** Attach velcro inner pads to the palm plate. Mount ESP32 #2 and MPU6050 securely inside the upper forearm casing.

PART 3: REAL-WORLD USABILITY & SYNERGY

Section 3: Project Usability & Combination Analysis

Dedicated Use Cases & Why Combining Both Guarantees Success

1. Usability: Haptic Communication System

Societal Accessibility Impact: Provides visually impaired users with real-time non-auditory 360° spatial awareness. By detecting head-height and body-height obstacles via the neckband and outputting tactile pulses to the sleeve, it acts as a reliable navigation safety net.

2. Usability: Mechanical Iron Man Gauntlet

Mechanical Prototyping & Control: Demonstrates fast-donning wearable mechanical engineering. Elastic linkages allow one-handed attachment over heavy gear, while the internal MPU6050 tracks wrist gestures for hands-free industrial control.

3. Why Combining Both Wins Hack Club Fallen ($1 + 1 = 3$)

- **Fail-Safe Design Strategy:** The Haptic Navigation System serves as the high-impact, functional core. Even if mechanical armor tweaks fail during a demo, the working assistive device secures top judge scores.
- **Mechanical Protection:** The gauntlet acts as a sleek mechanical shell protecting the sleeve box and wiring.
- **Unbeatable Demo Wow-Factor:** Combines real-world ethical utility with a thrilling mechanical kinetic showcase.

PART 4: 2-MEMBER TEAM EXECUTION

Section 4: 2-Member Team 1,000 XP Strategy

Task Allocation & 3-Sunday Sprint Log Strategy

1. 2-Member Task Allocation

Teammate	Primary Domain	Specific Build Responsibilities
Teammate 1 (You)	**Electronics & Firmware**	Neckband wiring, BNO085 compass integration, haptic motor code, ESP32 firmware.
Teammate 2	**Mechanical CAD & Rigging**	Linkage ridge 3D printing, elastic tensioner assembly, sleeve stitching, velcro mounting.

2. How to Hit 1,000 XP in 3 Sundays

- **Multiplier Rate:** Using character Sami ($1.75\times$ multiplier) = 35 XP/hour ($30\text{ hours} \times 35 = \mathbf{1,050 \text{ XP}}$).
- **The 3-Sunday Work Plan:** Work 10 physical build hours on 3 consecutive Sundays.
- **Daily Journal Distribution (Crucial):** Divide your 10-hour Sunday logs into 5 separate daily journal posts (2 hours/day, Sunday–Thursday). This prevents hitting daily caps, maintains your 5-day streak bonus (+30 XP/week), and guarantees 1,000+ XP in 3 weeks!

> **Shenzhen Submission Checklist:**
> Ensure your KiCad schematics are clean, maintain your daily journaling streaks, and record a video of the full mechanical deployment!